

Woojin Cho

+82 10-8225-8229 | f5foehn@snu.ac.kr

 [WoojinCho-Ryan](#)

Seoul, South Korea

PUBLICATIONS

- **K-WebArena: A Challenging Web Navigation Agent Benchmark on Korean Websites** *Under Preparation*
 - Woojin Cho, Dongkeun Yoon, Guijin Son, Nahyun Lee, Seungone Kim
- **K-BrowseComp: A Web Browsing Agent Benchmark Grounded in Korean Contexts** *Preprint, May 2026*
 - Nahyun Lee, Dongkeun Yoon, Guijin Son, . . . , Woojin Cho, Seungone Kim
 - [arXiv:2606.02404](#)

EDUCATION

- **Seoul National University** *Mar 2024 – Present*
B.S. in Computer Science and Engineering (Minor in Mathematical Sciences) *Seoul, South Korea*
 - **GPA: 4.06 / 4.30, Major GPA: 4.20 / 4.30**
 - Expected Graduation: Aug 2029 (Mandatory Military Service: Jan 2027 – Jun 2028)
- **Hansung Science High School** *Mar 2021 – Feb 2024*
Specialized in Mathematics and Science *Seoul, South Korea*

RELEVANT COURSEWORK

- **Computer Science:** Advanced Algorithms, Data Structures, Discrete Mathematics, Logic Design
- **Mathematics:** Linear Algebra I, Analysis I, Differential Equations, Abstract Algebra I

RESEARCH INTERESTS

- **Primary:** Mechanistic Interpretability, Autoformalization, Agentic Benchmarks

RESEARCH EXPERIENCE

- **Research Intern Mentorship** *Feb 2026 – Current*
Mentor: Seungone Kim
 - **Project: Korean Web Navigation Benchmark**
 - **Korean Web Environment Simulation:** Constructed reproducible Docker-based web environments for major Korean services (e.g., Naver, KTX) to enable realistic evaluation of web navigation agents.
 - **Benchmark Localization:** Curated a diverse set of Korea-specific navigation tasks and queries, addressing the cultural and linguistic gaps in existing English-centric benchmarks like WebArena.
 - **Framework Adaptation & Scalability:** Adapted the WebArena framework to handle Korean web interfaces and designed the architecture to support future extensions into Computer Use Agents (CUA).
- **Language & Data Intelligence Lab, Seoul National University** *Sep 2025 – Jan 2026*
Undergraduate Research Intern (Advisor: Prof. Seung-won Hwang) *Seoul, South Korea*
 - **Project: Autoformalization & Structured Reasoning**
 - **Designing Intermediate Representations:** Designed a structured JSON schema to bridge the semantic gap between ambiguous natural language math problems and formal logic (Lean 4), enabling rigorous verification.
 - **Data Pipeline Engineering:** Built a robust preprocessing pipeline to convert unstructured mathematical texts into structured evaluation datasets, handling diverse edge cases in logical statements.
 - **Hallucination Analysis:** Analyzed failure modes of LLMs in reasoning tasks, identifying that enforcing output structure significantly mitigates logical hallucinations.

PROJECTS

- **Adaptive Robustness of CountSketch: Collision Noise & Superposition**

Spring 2026

Theory, Randomized Algorithms, Mechanistic Interpretability

- Proved a complete non-adaptive analysis of CountSketch — unbiasedness, variance bound, and median amplification — isolating the single independence hypothesis each proof relies on.
- Derived a **conditional-expectation lemma** showing that adaptive query selection converts mean-zero collision noise into biased noise, connecting the mechanism to the Hardt–Woodruff impossibility theorem.
- Proposed a **formal dictionary** between CountSketch and toy models of superposition, framing adaptive fragility as a precise toy model of an interpretability hazard in compressed neural representations.

- **Real-time Visual Perception & Perspective Correction**

Spring 2024

Python, NumPy, Linear Algebra

- Implemented a **real-time vision algorithm** to rectify distorted monitor screens (e.g., trapezoids) into canonical rectangular views using **Homography** matrix operations.
- Applied **Linear Algebra** principles to map pixel coordinates from a tilted camera feed to a target 2D plane, correcting geometric distortions.
- Optimized the pipeline for video streams, developing robustness against camera jitter and motion blur to ensure stable detection.

SKILLS

- **Languages:** Python (Advanced), C++, Java, Lean 4

- **Frameworks & Libraries:** PyTorch, Hugging Face Transformers, NumPy, Pandas

- **Tools & DevOps:** GitHub, Linux (Ubuntu), $\text{\LaTeX}$